

Quantum Mechanics

Course notes

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1 Waves and Particles. Introduction to the Basic Ideas of Quantum Mechanics

1.A Electromagnetic Waves and Photons

The interaction of an electromagnetic wave with matter occurs by means of **elementary indivisible processes**, in which the radiation appears to be composed of particles, the **photons**. Their energy and momentum are given by the **Planck–Einstein relations**

$$E = h\nu = \hbar\omega, \quad \vec{p} = \hbar\vec{k},$$

where $h \simeq 6.62 \times 10^{-34}$ J s is the Planck constant and $\hbar := h/2\pi$.

1.B Material Particles and Matter Waves

The energy of an atom is **quantised**: it can only take certain discrete values E_i ($i = 1, 2, \dots, n$). The emission or absorption of a photon is then accompanied by a “jump” in the energy of the atom from one permitted value E_i to another E_j , and the energy of the photon is

$$h\nu_{ij} = |E_i - E_j|.$$

De Broglie’s hypothesis: material particles, just like photons, can have a wave-like aspect. As a consequence, interference patterns can be obtained with material particles such as electrons. One therefore associates with a material particle of energy E and momentum $\vec{p}$ a wave whose angular frequency $\omega = 2\pi\nu$ and wave vector $\vec{k}$ are given by the same relations as for photons,

$$E = h\nu = \hbar\omega, \quad \vec{p} = \hbar\vec{k},$$

which leads to the **de Broglie relation**

$$\lambda := \frac{2\pi}{|\vec{k}|} = \frac{h}{|\vec{p}|}.$$

For a particle of mass m in a potential $V(\vec{r}, t)$, the **Schrödinger equation** takes the form

$$i\hbar \partial_t \Psi(\vec{r}, t) = -\frac{\hbar^2}{2m} \Delta \Psi(\vec{r}, t) + V(\vec{r}, t) \Psi(\vec{r}, t). \quad (1.1)$$

The probability of finding the particle in d^3r is

$$d\mathcal{P}(\vec{r}, t) = C |\Psi(\vec{r}, t)|^2 d^3r, \quad \int_{\mathbb{R}^3} d\mathcal{P}(\vec{r}, t) = 1.$$

1.C Quantum Description of a Particle. Wave Packets

Let us analyse the free particle, $V(\vec{r}, t) = 0$:

$$i\hbar \partial_t \Psi(\vec{r}, t) = -\frac{\hbar^2}{2m} \Delta \Psi(\vec{r}, t).$$

With the ansatz $\Psi(\vec{r}, t) = A e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ we get

$$\hbar\omega = \frac{\hbar^2 |\vec{k}|^2}{2m} \implies \omega = \frac{\hbar |\vec{k}|^2}{2m},$$

which, by de Broglie's hypothesis, is $E = p^2/2m$. Since $|\Psi(\vec{r}, t)|^2 = |A|^2$, the probability of presence is uniform throughout space.

Due to the **superposition principle**, a general solution is the **three-dimensional wave packet**

$$\Psi(\vec{r}, t) = \frac{1}{(2\pi)^{3/2}} \int d^3k \, g(\vec{k}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}.$$

In one dimension and at $t = 0$, ψ and g are Fourier transforms of each other:

$$\psi(x, 0) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} dk \, g(k) e^{ikx}, \quad g(k) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} dx \, \psi(x, 0) e^{-ikx}.$$

At a given instant a wave packet might look like Fig. 1.1.

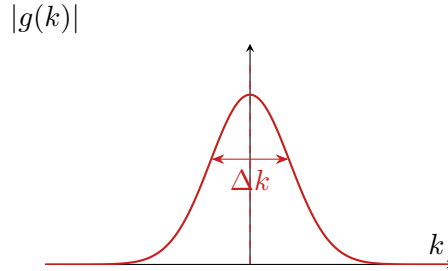


Figure 1.1: Modulus of the amplitude $g(k)$, centred at k_0 with width Δk .

A simple model. Let us assume that such a wave packet is the sum of three plane waves:

$$\begin{aligned} \psi(x) &= \frac{g(k_0)}{\sqrt{2\pi}} \left[e^{ik_0 x} + \frac{1}{2} e^{i(k_0 - \frac{\Delta k}{2})x} + \frac{1}{2} e^{i(k_0 + \frac{\Delta k}{2})x} \right] \\ &= \frac{g(k_0)}{\sqrt{2\pi}} e^{ik_0 x} \left[1 + \cos\left(\frac{\Delta k}{2} x\right) \right]. \end{aligned}$$

The interference becomes destructive when $1 + \cos(\cdot) = 0$, i.e. when the argument equals $\pm\pi$; this is used to define Δx as the distance between the two zeros of $|\psi(x)|$:

$$\frac{\Delta k}{2} \frac{\Delta x}{2} := \pi \quad \implies \quad \Delta k \Delta x = 4\pi.$$

The smaller the width Δk of $|g(k)|$, the larger the width Δx of $\psi(x)$.

General case. Consider now $g(k) = |g(k)| e^{i\alpha(k)}$, assuming that $\alpha(k)$ varies slowly within the interval $[k_0 - \Delta k/2, k_0 + \Delta k/2]$ where $|g(k)|$ is appreciable. Taking Δk sufficiently small, one can expand $\alpha(k)$ in the neighbourhood of $k = k_0$:

$$\alpha(k) \simeq \alpha(k_0) + (k - k_0) \left. \frac{d\alpha}{dk} \right|_{k_0}, \quad x_0 := - \left. \frac{d\alpha}{dk} \right|_{k_0}.$$

Then

$$\begin{aligned} \psi(x) &= \frac{1}{\sqrt{2\pi}} \int dk \, |g(k)| e^{i\alpha(k)} e^{ikx} \simeq \frac{1}{\sqrt{2\pi}} \int dk \, |g(k)| e^{i[\alpha(k_0) - x_0(k - k_0)]} e^{ikx} \\ &= \frac{e^{i(k_0 x + \alpha(k_0))}}{\sqrt{2\pi}} \int dk \, |g(k)| e^{-ix_0(k - k_0)} e^{i(k - k_0)x}, \end{aligned}$$

that is,

$$\psi(x) \simeq \frac{e^{i(k_0 x + \alpha(k_0))}}{\sqrt{2\pi}} \int dk |g(k)| e^{i(x-x_0)(k-k_0)}. \quad (1.2)$$

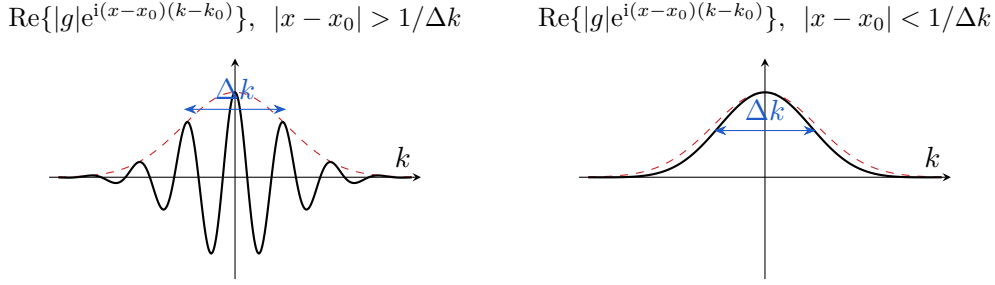


Figure 1.2: Integrand of (1.2) with its envelope $|g(k)|$ (dashed). Left: far from x_0 the integrand oscillates many times over Δk and $|\psi(x)|$ is small. Right: for x close to x_0 it does not oscillate and $|\psi(x)|$ is maximal.

When x moves away from x_0 , $|\psi(x)|$ decreases. The decrease becomes appreciable if $e^{i(k-k_0)(x-x_0)}$ oscillates approximately once when k traverses the domain Δk , that is, when

$$\Delta k (x - x_0) \simeq 1.$$

If Δx is the approximate width of the wave packet, then

$$\Delta k \Delta x \gtrsim 1,$$

which is a general property of waves (nothing quantum yet). Taking into account the de Broglie relation $p = \hbar k$, however,

$$\Delta x \Delta p \gtrsim \hbar,$$

which is a quantum statement because of the de Broglie relation.

Time evolution of a free wave packet. The motion of a free wave packet is not given by the phase velocities of the individual plane waves. For a plane wave $e^{i(kx - \omega t)}$ the **phase velocity** is

$$v_p = \frac{\omega}{k}.$$

The velocity of propagation of the wave packet, however, is the **group velocity**

$$v_g = \frac{d\omega}{dk}.$$

To see this, write

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int dk |g(k)| e^{i\alpha(k)} e^{i(kx - \omega(k)t)}$$

and expand, with $\alpha_0 = \alpha(k_0)$, $\omega_0 = \omega(k_0)$,

$$\alpha(k) \simeq \alpha_0 + \left. \frac{d\alpha}{dk} \right|_{k_0} (k - k_0) = \alpha_0 - x_0(k - k_0), \quad \omega(k) \simeq \omega_0 + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0).$$

Then

$$\begin{aligned} \psi(x, t) &\simeq \frac{e^{i(\alpha_0 - \omega_0 t + k_0 x)}}{\sqrt{2\pi}} \int dk |g(k)| e^{-i(x_0 + \omega'(k_0)t)(k - k_0)} e^{i(k - k_0)x} \\ &= \frac{e^{i(\alpha_0 - \omega_0 t + k_0 x)}}{\sqrt{2\pi}} \int dk |g(k)| e^{i[(x - x_0) - \omega'(k_0)t](k - k_0)}. \end{aligned}$$

The maximum is located where the square bracket vanishes,

$$x_M(t) = x_0 + \left. \frac{d\omega}{dk} \right|_{k_0} t,$$

so the maximum of the wave packet moves with velocity $v_g = d\omega/dk$.

1.D Particle in a Time-Independent Scalar Potential

Separation of variables. Stationary states. The Schrödinger equation for a time-independent potential is

$$i\hbar \partial_t \Psi(\vec{r}, t) = -\frac{\hbar^2}{2m} \Delta \Psi(\vec{r}, t) + V(\vec{r}) \Psi(\vec{r}, t).$$

We look for **stationary states** of the form $\Psi(\vec{r}, t) = \varphi(\vec{r}) \chi(t)$, which gives

$$\frac{i\hbar \partial_t \chi}{\chi} = -\frac{\hbar^2}{2m} \frac{\Delta \varphi}{\varphi} + V(\vec{r}).$$

Both sides must equal a constant, $\hbar\omega$:

$$\begin{aligned} \frac{i\hbar \partial_t \chi}{\chi} = \hbar\omega &\implies \chi(t) = A e^{-i\omega t}, \\ -\frac{\hbar^2}{2m} \Delta \varphi + V(\vec{r}) \varphi = \hbar\omega \varphi, &\quad \Psi(\vec{r}, t) = \varphi(\vec{r}) e^{-i\omega t}. \end{aligned}$$

This is a **stationary solution** of the Schrödinger equation: it is a state with well-defined energy $E = \hbar\omega$ and a time-independent probability density, $|\Psi(\vec{r}, t)|^2 = |\varphi(\vec{r})|^2$.

The equation for φ reads

$$\left[-\frac{\hbar^2}{2m} \Delta + V(\vec{r}) \right] \varphi(\vec{r}) = E \varphi(\vec{r}),$$

where the (linear) operator in brackets defines the **Hamiltonian** H :

$$H \varphi(\vec{r}) = E \varphi(\vec{r}).$$

Since the Schrödinger equation is linear, if there exist solutions

$$H \varphi_n(\vec{r}) = E_n \varphi_n(\vec{r}),$$

then $\Psi_n(\vec{r}, t) = \varphi_n(\vec{r}) e^{-iE_n t/\hbar}$, and the general solution is

$$\Psi(\vec{r}, t) = \sum_n c_n \varphi_n(\vec{r}) e^{-iE_n t/\hbar},$$

with arbitrary complex constants c_n .

2 The Mathematical Tools of Quantum Mechanics

2.A Space of the One-Particle Wave Function

Since $\int d^3r |\psi(\vec{r}, t)|^2 = 1$, where ψ satisfies the Schrödinger equation, we are led to study the set of square-integrable functions:

$$\psi \in \mathcal{F} \subset L^2,$$

which is a vector space.

Scalar product. If $\varphi, \psi \in \mathcal{F}$,

$$(\varphi, \psi) := \int d^3r \varphi^*(\vec{r}) \psi(\vec{r}), \quad (\varphi, \psi) = (\psi, \varphi)^*.$$

Commutator. Let A and B be operators. Their commutator is

$$[A, B] := AB - BA.$$

Discrete orthonormal basis in $\mathcal{F}$: $\{u_i(\vec{r})\}$. Consider a countable set of functions $u_i(\vec{r}) \in \mathcal{F}$, $i = 1, 2, \dots$. The set $\{u_i(\vec{r})\}$ is **orthonormal** if

$$(u_i, u_j) = \int d^3r u_i^*(\vec{r}) u_j(\vec{r}) = \delta_{ij},$$

and it constitutes a **basis** if every function in $\mathcal{F}$ can be expanded in one and only one way:

$$\psi \in \mathcal{F} : \quad \psi(\vec{r}) = \sum_i c_i u_i(\vec{r}).$$

Notice that

$$(u_j, \psi) = \sum_i c_i (u_j, u_i) = c_j \implies c_i = (u_i, \psi) = \int d^3r u_i^*(\vec{r}) \psi(\vec{r}).$$

Closure relation.

$$\begin{aligned} \psi(\vec{r}) &= \sum_i c_i u_i(\vec{r}) = \sum_i (u_i, \psi) u_i(\vec{r}) \\ &= \sum_i \int d^3r' u_i^*(\vec{r}') \psi(\vec{r}') u_i(\vec{r}) = \int d^3r' \psi(\vec{r}') \sum_i u_i^*(\vec{r}') u_i(\vec{r}), \end{aligned}$$

hence the **closure relation**

$$\sum_i u_i^*(\vec{r}') u_i(\vec{r}) = \delta(\vec{r} - \vec{r}').$$

Bases not belonging to $\mathcal{F}$. *Plane waves.* In one dimension,

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} dp \bar{\psi}(p) e^{ipx/\hbar}, \quad \bar{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} dx \psi(x) e^{-ipx/\hbar}.$$

Defining

$$v_p(x) := \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \quad v_p \notin \mathcal{F},$$

we have

$$\psi(x) = \int_{\mathbb{R}} dp \bar{\psi}(p) v_p(x), \quad \bar{\psi}(p) = (v_p, \psi) = \int dx v_p^*(x) \psi(x),$$

with

$$\begin{aligned} \int_{\mathbb{R}} dp v_p(x) v_p^*(x') &= \frac{1}{2\pi} \int \frac{dp}{\hbar} e^{i\frac{p}{\hbar}(x-x')} = \delta(x - x') && \text{(closure),} \\ (v_p, v_{p'}) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{dx}{\hbar} e^{i\frac{x}{\hbar}(p'-p)} = \delta(p - p') && \text{(orthonormality).} \end{aligned}$$

In three dimensions,

$$v_{\vec{p}}(\vec{r}) = \left(\frac{1}{2\pi\hbar}\right)^{3/2} e^{i\vec{p}\cdot\vec{r}/\hbar}, \quad \psi(\vec{r}) = \int d^3p \, \bar{\psi}(\vec{p}) v_{\vec{p}}(\vec{r}),$$

and the scalar product can be computed in either variable:

$$\begin{aligned} (\varphi, \psi) &= \int d^3r \, \varphi^*(\vec{r}) \psi(\vec{r}) = \int d^3r \int d^3p \int d^3p' \, \bar{\varphi}^*(\vec{p}) v_{\vec{p}}^*(\vec{r}) \bar{\psi}(\vec{p}') v_{\vec{p}'}(\vec{r}) \\ &= \int d^3p \int d^3p' \, \delta(\vec{p} - \vec{p}') \bar{\varphi}^*(\vec{p}) \bar{\psi}(\vec{p}') = \int d^3p \, \bar{\varphi}^*(\vec{p}) \bar{\psi}(\vec{p}). \end{aligned}$$

Delta functions. Define

$$\xi_{\vec{r}_0}(\vec{r}) = \delta(\vec{r} - \vec{r}_0) \notin \mathcal{F}.$$

Then

$$\begin{aligned} \psi(\vec{r}) &= \int d^3r_0 \, \psi(\vec{r}_0) \xi_{\vec{r}_0}(\vec{r}), \quad \psi(\vec{r}_0) = \int d^3r \, \xi_{\vec{r}_0}^*(\vec{r}) \psi(\vec{r}) = (\xi_{\vec{r}_0}, \psi), \\ (\varphi, \psi) &= \int d^3r_0 \, \varphi^*(\vec{r}_0) \psi(\vec{r}_0), \end{aligned}$$

and

$$\begin{aligned} \int d^3r_0 \, \xi_{\vec{r}_0}(\vec{r}) \xi_{\vec{r}_0}^*(\vec{r}') &= \int d^3r_0 \, \delta(\vec{r} - \vec{r}_0) \delta(\vec{r}' - \vec{r}_0) = \delta(\vec{r} - \vec{r}') \quad (\text{closure}), \\ (\xi_{\vec{r}_0}, \xi_{\vec{r}'_0}) &= \int d^3r \, \xi_{\vec{r}_0}^*(\vec{r}) \xi_{\vec{r}'_0}(\vec{r}) = \delta(\vec{r}_0 - \vec{r}'_0) \quad (\text{orthonormality}). \end{aligned}$$

Later we will write $v_{\vec{p}} \rightarrow |\vec{p}\rangle$ and $\xi_{\vec{r}} \rightarrow |\vec{r}\rangle$.

2.B State Space. Dirac Notation

Each quantum state of a particle is characterised by a **state vector** belonging to an abstract space $\mathcal{E}$, called the **state space** of a particle (a Hilbert space).

Any element of $\mathcal{E}$ is called a **ket** vector and is represented by $|\cdot\rangle$, e.g. $|\psi\rangle$. The wave function $\psi(\vec{r})$ gives the components of $|\psi\rangle$ in the $\vec{r}$ basis.

Any element of the dual space $\mathcal{E}^*$ is called a **bra** vector and is symbolised by $\langle\cdot|$. For example, the bra $\langle\chi|$ designates the linear functional χ . We shall henceforth use the notation $\langle\chi|\psi\rangle$ to denote the number obtained by causing the linear functional $\langle\chi| \in \mathcal{E}^*$ to act on the ket $|\psi\rangle \in \mathcal{E}$:

$$\chi(|\psi\rangle) = \langle\chi|\psi\rangle.$$

Correspondence between kets and bras. To every ket corresponds a bra, with $(|\varphi\rangle, |\psi\rangle) = \langle\varphi|\psi\rangle$. The correspondence is **antilinear**:

$$\lambda_1 |\varphi_1\rangle + \lambda_2 |\varphi_2\rangle \quad \longleftrightarrow \quad \lambda_1^* \langle\varphi_1| + \lambda_2^* \langle\varphi_2|.$$

However, **not to every bra corresponds a ket**.

Projectors. P is a **projector** if $P^2 = P$. Examples:

- projector onto $|\psi\rangle$: $P_\psi = |\psi\rangle\langle\psi|$;
- projector onto a q -dimensional subspace spanned by orthonormal $|\varphi_i\rangle$: $P_q = \sum_{i=1}^q |\varphi_i\rangle\langle\varphi_i|$.

Hermitian conjugation.

$$|\psi'\rangle = A|\psi\rangle \iff \langle\psi'| = \langle\psi|A^\dagger,$$

where $A^\dagger$ is the **adjoint operator** of A . Since $\langle\psi'|\varphi\rangle = \langle\varphi|\psi'\rangle^*$,

$$\langle\psi|A^\dagger|\varphi\rangle = \langle\varphi|A|\psi\rangle^*.$$

Other properties:

$$(A^\dagger)^\dagger = A, \quad (\lambda A)^\dagger = \lambda^* A^\dagger, \quad (A+B)^\dagger = A^\dagger + B^\dagger, \quad (AB)^\dagger = B^\dagger A^\dagger, \quad (|u\rangle\langle v|)^\dagger = |v\rangle\langle u|.$$

If the operator A is equal to its adjoint, $A = A^\dagger$ (i.e. $\langle\psi|A|\varphi\rangle = \langle\varphi|A|\psi\rangle^*$), then A is called **Hermitian**.

2.D Eigenvalue Equations. Observables

If

$$A|\psi\rangle = \lambda|\psi\rangle,$$

$|\psi\rangle$ is an **eigenvector** of A with **eigenvalue** λ . If there is only one eigenvector (up to a factor) for an eigenvalue, the eigenvalue is **non-degenerate** (simple). If there is more than one linearly independent eigenvector for an eigenvalue, it is **degenerate**, and the number of independent eigenvectors is the **degree of degeneracy**.

Observables. Let A be Hermitian, $A^\dagger = A$.

- *The eigenvalues of a Hermitian operator are real.* If $A|\psi\rangle = \lambda|\psi\rangle$ with $\langle\psi|\psi\rangle = 1$, then $\langle\psi|A|\psi\rangle = \lambda$. Taking the complex conjugate, $\langle\psi|A^\dagger|\psi\rangle = \lambda^*$, and since $A^\dagger = A$, $\lambda = \lambda^* \in \mathbb{R}$.
- *Two eigenvectors of a Hermitian operator corresponding to two different eigenvalues are orthogonal.* If $A|\psi\rangle = \lambda|\psi\rangle$ and $A|\varphi\rangle = \mu|\varphi\rangle$, then $\langle\varphi|A|\psi\rangle = \lambda\langle\varphi|\psi\rangle$ and $\langle\varphi|A|\psi\rangle = \mu\langle\varphi|\psi\rangle$ (acting to the left with real μ), so

$$0 = (\mu - \lambda)\langle\varphi|\psi\rangle \xrightarrow{\mu \neq \lambda} \langle\varphi|\psi\rangle = 0.$$

By definition, the Hermitian operator A is an **observable** if its orthonormal eigenvectors form a basis of the state space. Writing

$$A|\psi_n^i\rangle = a_n|\psi_n^i\rangle, \quad i = 1, \dots, g_n,$$

the eigenvectors of different eigenvalues are orthogonal, $\langle\psi_n^i|\psi_m^j\rangle = 0$ for $n \neq m$, and within each eigensubspace one can always choose $\langle\psi_n^i|\psi_n^j\rangle = \delta_{ij}$. Hence

$$\langle\psi_n^i|\psi_n^{i'}\rangle = \delta_{nn'}\delta_{ii'},$$

and A is an observable if the **closure relation**

$$\sum_{n=1}^{\infty} \sum_{i=1}^{g_n} |\psi_n^i\rangle\langle\psi_n^i| = \mathbb{1}$$

holds, i.e. if $\{|\psi_n^i\rangle\}$ is a basis of $\mathcal{E}$.

Sets of commuting observables.

Statement. If $[A, B] = 0$ and $A|\psi\rangle = a|\psi\rangle$, then $B|\psi\rangle$ is also an eigenvector of A with the same eigenvalue.

Indeed, if $|\psi\rangle \in \mathcal{E}_a$ (the eigensubspace of a), then $B|\psi\rangle \in \mathcal{E}_a$: $\mathcal{E}_a$ is globally invariant under the action of B .

Statement. If two observables commute, $[A, B] = 0$, one can construct an orthonormal basis of the state space with eigenvectors common to A and B .

Complete set of commuting observables (C.S.C.O.). If none of the eigenvalues of A is degenerate, specifying the eigenvalue determines the eigenvector uniquely, and A (by itself) is a C.S.C.O. If there is degeneracy, we can use another observable B with $[A, B] = 0$ to characterise the degenerate eigenvectors; if this removes the degeneracy, A and B are said to form a C.S.C.O. The procedure works for an arbitrary number of commuting observables.

2.E Two Important Representations

With

$$\xi_{\vec{r}_0}(\vec{r}) = \delta(\vec{r} - \vec{r}_0) \longleftrightarrow |\vec{r}_0\rangle, \quad v_{\vec{p}_0}(\vec{r}) = (2\pi\hbar)^{-3/2} e^{i\vec{p}_0 \cdot \vec{r}/\hbar} \longleftrightarrow |\vec{p}_0\rangle,$$

we have

$$\langle \vec{r}_0 | \psi \rangle = \psi(\vec{r}_0), \quad \langle \vec{p}_0 | \psi \rangle = \bar{\psi}(\vec{p}_0), \quad \langle \vec{r}_0 | \vec{p}_0 \rangle = v_{\vec{p}_0}(\vec{r}_0) = (2\pi\hbar)^{-3/2} e^{i\vec{p}_0 \cdot \vec{r}_0/\hbar}.$$

Notice that

$$\begin{aligned} \langle \vec{r} | P_x | \psi \rangle &= \int d^3p \langle \vec{r} | \vec{p} \rangle \langle \vec{p} | P_x | \psi \rangle = (2\pi\hbar)^{-3/2} \int d^3p e^{i\vec{p} \cdot \vec{r}/\hbar} p_x \bar{\psi}(\vec{p}) \\ &= \frac{\hbar}{i} \partial_x \left[(2\pi\hbar)^{-3/2} \int d^3p e^{i\vec{p} \cdot \vec{r}/\hbar} \bar{\psi}(\vec{p}) \right] = \frac{\hbar}{i} \partial_x \langle \vec{r} | \psi \rangle, \end{aligned}$$

so that

$$\langle \vec{r} | \vec{P} | \psi \rangle = -i\hbar \nabla \langle \vec{r} | \psi \rangle.$$

3 The Postulates of Quantum Mechanics

3.B Statement of the Postulates

1. At a fixed time t_0 , the state of an isolated physical system is defined by specifying a ket $|\psi(t_0)\rangle$ belonging to the state space $\mathcal{E}$.
2. Every measurable physical quantity $\mathcal{A}$ is described by an operator A acting in $\mathcal{E}$; this operator is an observable ($A^\dagger = A$ and its eigenvectors are a basis of $\mathcal{E}$).
3. The only possible result of the measurement of a physical quantity $\mathcal{A}$ is one of the eigenvalues of the corresponding observable A .
4. When the physical quantity $\mathcal{A}$ is measured on a system in the normalised state $|\psi\rangle$, the probability $\mathcal{P}(a_n)$ of obtaining the non-degenerate eigenvalue a_n of the corresponding observable A is

$$\mathcal{P}(a_n) = |\langle u_n | \psi \rangle|^2,$$

where $|u_n\rangle$ is the normalised eigenvector of A associated with the eigenvalue a_n . If a_n is g_n -fold degenerate,

$$\mathcal{P}(a_n) = \sum_{i=1}^{g_n} |\langle u_n^i | \psi \rangle|^2.$$

5. If the measurement of the physical quantity $\mathcal{A}$ on the system in the state $|\psi\rangle$ gives the result a_n , the state of the system immediately after the measurement is the normalised projection

$$\frac{P_n |\psi\rangle}{\sqrt{\langle\psi|P_n|\psi\rangle}}$$

of $|\psi\rangle$ onto the eigensubspace associated with a_n .

6. The time evolution of the state vector $|\psi(t)\rangle$ is governed by the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle, \quad (3.1)$$

where the Hamiltonian $H(t)$ is the observable associated with the total energy of the system.

Quantisation rule. The observable A which describes a classically defined physical quantity $\mathcal{A}$ is obtained by replacing, in the suitably symmetrised expression for $\mathcal{A}$, $\vec{r}$ and $\vec{p}$ by the observables $\vec{R}$ and $\vec{P}$.

3.C Physical Interpretation of the Postulates

C.4 Mean value of an observable

If $|\psi\rangle$ is normalised, the **mean value** of the observable A is

$$\langle A \rangle_\psi = \langle A \rangle := \langle \psi | A | \psi \rangle.$$

Inserting the closure relation,

$$\langle A \rangle = \langle \psi | A \sum_n \sum_{i=1}^{g_n} |u_n^i\rangle \langle u_n^i| | \psi \rangle = \sum_n \sum_{i=1}^{g_n} a_n \langle \psi | u_n^i \rangle \langle u_n^i | \psi \rangle = \sum_n \sum_{i=1}^{g_n} a_n \left| \langle u_n^i | \psi \rangle \right|^2,$$

that is,

$$\langle A \rangle = \sum_n a_n \mathcal{P}(a_n).$$

C.5 The root-mean-square deviation

$$\Delta A := \sqrt{\langle (A - \langle A \rangle)^2 \rangle},$$

$$(\Delta A)^2 = \langle A^2 - 2A \langle A \rangle + \langle A \rangle^2 \rangle = \langle A^2 \rangle - 2 \langle A \rangle^2 + \langle A \rangle^2,$$

so that

$$(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2.$$

With this definition of ΔA the lower bound of the **Heisenberg relation** is well defined:

$$\Delta X_i \Delta P_i \geq \frac{\hbar}{2}.$$

3.D The Physical Implications of the Schrödinger Equation

Linearity. If $|\psi_1\rangle$ and $|\psi_2\rangle$ are solutions, then $\lambda_1 |\psi_1\rangle + \lambda_2 |\psi_2\rangle$ is also a solution.

Conservation of probability. Let $\langle \psi(t_0) | \psi(t_0) \rangle = 1$. Using (3.1) and its adjoint, $-\mathrm{i}\hbar \frac{d}{dt} \langle \psi | = \langle \psi | H$,

$$\frac{d}{dt} \langle \psi(t) | \psi(t) \rangle = \left(\frac{d}{dt} \langle \psi | \right) | \psi \rangle + \langle \psi | \left(\frac{d}{dt} | \psi \rangle \right) = -\frac{1}{\mathrm{i}\hbar} \langle \psi | H | \psi \rangle + \frac{1}{\mathrm{i}\hbar} \langle \psi | H | \psi \rangle = 0,$$

so

$$\langle \psi(t) | \psi(t) \rangle = \langle \psi(t_0) | \psi(t_0) \rangle = 1.$$

Local conservation of probability. Probability densities and probability currents.
The **probability density** is

$$\rho(\vec{r}, t) := |\psi(\vec{r}, t)|^2.$$

With $H = \vec{P}^2/2m + V(\vec{R}, t)$,

$$\begin{aligned} \mathrm{i}\hbar \partial_t \psi &= -\frac{\hbar^2}{2m} \Delta \psi + V \psi, \\ -\mathrm{i}\hbar \partial_t \psi^* &= -\frac{\hbar^2}{2m} \Delta \psi^* + V \psi^*, \end{aligned}$$

and multiplying the first by ψ^* , the second by ψ and subtracting,

$$\mathrm{i}\hbar \partial_t (\psi \psi^*) = -\frac{\hbar^2}{2m} [\psi^* \Delta \psi - \psi \Delta \psi^*],$$

$$\frac{\partial \rho}{\partial t} + \frac{\hbar}{2m\mathrm{i}} [\psi^* \Delta \psi - \psi \Delta \psi^*] = 0 \quad \implies \quad \frac{\partial \rho}{\partial t} + \nabla \cdot \left[\frac{\hbar}{2m\mathrm{i}} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right] = 0.$$

The **probability current** is

$$\vec{J} := \frac{\hbar}{2m\mathrm{i}} (\psi^* \nabla \psi - \psi \nabla \psi^*) = \frac{1}{m} \operatorname{Re} \left[\psi^* \frac{\hbar}{\mathrm{i}} \nabla \psi \right],$$

and

$$\partial_t \rho + \nabla \cdot \vec{J} = 0.$$

Evolution of the mean value of an observable; relationship with classical mechanics.

Differentiating $\langle A \rangle(t) = \langle \psi(t) | A(t) | \psi(t) \rangle$,

$$\begin{aligned} \frac{d}{dt} \langle A \rangle &= \left(\frac{d}{dt} \langle \psi | \right) A | \psi \rangle + \langle \psi | A \left(\frac{d}{dt} | \psi \rangle \right) + \langle \psi | \frac{\partial A}{\partial t} | \psi \rangle \\ &= -\frac{1}{\mathrm{i}\hbar} \langle \psi | H A | \psi \rangle + \frac{1}{\mathrm{i}\hbar} \langle \psi | A H | \psi \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle = \frac{\mathrm{i}}{\hbar} \langle [H, A] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle, \end{aligned}$$

that is,

$$\frac{d}{dt} \langle A \rangle = \frac{1}{\mathrm{i}\hbar} \langle [A, H] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle.$$

Application to $\vec{R}$ and $\vec{P}$, with $[R_i, P_j] = \mathrm{i}\hbar \delta_{ij}$ and $H = \vec{P}^2/2m + V(\vec{R})$:

$$\begin{aligned} \frac{d}{dt} \langle \vec{R} \rangle &= \frac{1}{\mathrm{i}\hbar} \langle [\vec{R}, H] \rangle = \frac{1}{\mathrm{i}\hbar} \left\langle \left[\vec{R}, \frac{\vec{P}^2}{2m} \right] \right\rangle = \frac{\langle \vec{P} \rangle}{m}, \\ \frac{d}{dt} \langle \vec{P} \rangle &= \frac{1}{\mathrm{i}\hbar} \langle [\vec{P}, H] \rangle = \frac{1}{\mathrm{i}\hbar} \langle [\vec{P}, V(\vec{R})] \rangle = -\langle \nabla V(\vec{R}) \rangle, \end{aligned}$$

where $[\vec{P}, V(\vec{R})] = -i\hbar \nabla V(\vec{R})$. These are **Ehrenfest's theorem**:

$$\frac{d}{dt} \langle \vec{R} \rangle = \frac{1}{m} \langle \vec{P} \rangle, \quad \frac{d}{dt} \langle \vec{P} \rangle = - \langle \nabla V(\vec{R}) \rangle.$$

They are similar to their classical counterparts, but they do not give the same result as the classical equations for the centre of the wave packet, because in general

$$\langle \nabla V(\vec{R}) \rangle \neq \nabla V(\vec{r})|_{\vec{r}=\langle \vec{R} \rangle}.$$

D.2 The case of conservative systems

Let us consider as basis the eigenvectors of the Hamiltonian of a **conservative system**, $H \neq H(t)$ (so $E_n \neq E_n(t)$):

$$H |\varphi_{n,\tau}\rangle = E_n |\varphi_{n,\tau}\rangle,$$

where τ labels the degenerate eigenvectors. Then

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t) |\varphi_{n,\tau}\rangle, \quad c_{n,\tau}(t) = \langle \varphi_{n,\tau} | \psi(t) \rangle,$$

and

$$i\hbar \frac{d}{dt} \langle \varphi_{n,\tau} | \psi \rangle = \langle \varphi_{n,\tau} | H | \psi \rangle = E_n \langle \varphi_{n,\tau} | \psi \rangle \implies c_{n,\tau}(t) = c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar}.$$

Stationary states. If $|\psi(t_0)\rangle$ is an eigenstate of H with eigenvalue E_n ,

$$\begin{aligned} |\psi(t_0)\rangle &= \sum_{\tau} c_{n,\tau}(t_0) |\varphi_{n,\tau}\rangle \implies \\ |\psi(t)\rangle &= \sum_{\tau} c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar} |\varphi_{n,\tau}\rangle = e^{-iE_n(t-t_0)/\hbar} |\psi(t_0)\rangle. \end{aligned}$$

This is the same state up to a global phase factor: an eigenstate of H does not vary in time, and is called a **stationary state**.

Bohr frequencies of a system. Selection rules. Let B be an arbitrary observable; then

$$\frac{d}{dt} \langle B \rangle = \frac{1}{i\hbar} \langle [B, H] \rangle + \left\langle \frac{\partial B}{\partial t} \right\rangle.$$

For a conservative system we know $|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar} |\varphi_{n,\tau}\rangle$, so $\langle B \rangle(t)$ can be computed explicitly:

$$\langle \psi(t) | B | \psi(t) \rangle = \sum_{n,\tau} \sum_{n',\tau'} c_{n,\tau}^*(t_0) c_{n',\tau'}(t_0) e^{i(E_n - E_{n'})(t-t_0)/\hbar} \langle \varphi_{n,\tau} | B | \varphi_{n',\tau'} \rangle.$$

If $B \neq B(t)$, the matrix elements $\langle \varphi_{n,\tau} | B | \varphi_{n',\tau'} \rangle$ do not depend on time. Therefore the evolution of $\langle B \rangle(t)$ is described by a series of oscillating terms whose frequencies

$$\nu_{nn'} = \frac{1}{2\pi} \frac{|E_n - E_{n'}|}{\hbar} = \frac{|E_n - E_{n'}|}{h},$$

the **Bohr frequencies**, are characteristic of the system under consideration, but independent of B and of the initial state of the system. The weights, on the other hand, depend on B and on the initial state. The study of the non-diagonal elements $\langle \varphi_{n,\tau} | B | \varphi_{n',\tau'} \rangle$ gives rise to the **selection rules**, which indicate which frequencies can be emitted or absorbed under certain conditions.

The time–energy uncertainty relation.

$$\Delta t \Delta E \gtrsim \hbar.$$

It is not a relation of the same kind as the Heisenberg relation: there is no operator for time t .

Complement F_{III}: The Evolution Operator

The **evolution operator** $U(t, t_0)$ is defined by

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle, \quad U(t_0, t_0) = \mathbb{1}.$$

Substituting into (3.1),

$$i\hbar \frac{\partial}{\partial t} (U(t, t_0) |\psi(t_0)\rangle) = H(t) U(t, t_0) |\psi(t_0)\rangle \implies i\hbar \frac{\partial}{\partial t} U(t, t_0) = H(t) U(t, t_0),$$

or, in integral form,

$$U(t, t_0) = \mathbb{1} - \frac{i}{\hbar} \int_{t_0}^t dt' H(t') U(t', t_0).$$

For an infinitesimal time step,

$$d|\psi(t)\rangle = |\psi(t+dt)\rangle - |\psi(t)\rangle = -\frac{i}{\hbar} H(t) |\psi(t)\rangle dt \implies |\psi(t+dt)\rangle = \left(\mathbb{1} - \frac{i}{\hbar} H dt \right) |\psi(t)\rangle,$$

so

$$U(t+dt, t) = \mathbb{1} - \frac{i}{\hbar} H(t) dt :$$

H is the **generator of time evolution**. U is **unitary**, $U^\dagger(t, t') = U(t', t)$, so it preserves the norm of vectors.

If $H \neq H(t)$ (conservative system),

$$U(t, t_0) = e^{-iH(t-t_0)/\hbar}.$$

Complement G_{III}: The Schrödinger and Heisenberg Pictures

States	Operators	Picture
evolve	no evolution	Schrödinger
no evolution	evolve	Heisenberg

The Heisenberg state and operators are

$$|\psi_H\rangle = U^\dagger(t, t_0) |\psi_S(t)\rangle = U^\dagger(t, t_0) U(t, t_0) |\psi_S(t_0)\rangle = |\psi_S(t_0)\rangle,$$

$$A_H(t) = U^\dagger(t, t_0) A_S(t) U(t, t_0).$$

Differentiating, and inserting $UU^\dagger = \mathbb{1}$,

$$\frac{dA_H}{dt} = -\frac{1}{i\hbar} U^\dagger(t, t_0) H_S(t) A_S(t) U(t, t_0) + \frac{1}{i\hbar} U^\dagger(t, t_0) A_S(t) H_S(t) U(t, t_0) + U^\dagger(t, t_0) \frac{dA_S}{dt} U(t, t_0),$$

which gives the **Heisenberg equation of motion**

$$i\hbar \frac{dA_H}{dt} = [A_H, H_H] + i\hbar \left(\frac{dA_S}{dt} \right)_H.$$

4 Application of the Postulates to Simple Cases: Spin 1/2 and Two-Level Systems

This chapter has a title page only in the handwritten notes.

5 The One-Dimensional Harmonic Oscillator

5.A Introduction

General properties of the quantum mechanical Hamiltonian. In quantum mechanics $[X, P] = i\hbar$, and

$$H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2,$$

which does not depend on time. The eigenvalue equation $H|\varphi\rangle = E|\varphi\rangle$ reads, in the $\{|x\rangle\}$ representation,

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2 \right] \varphi(x) = E \varphi(x).$$

- The eigenvalues of H are positive, since the potential is bounded from below (by 0).
- The eigenfunctions of H have a definite parity, since $V(-x) = V(x)$.
- The energy spectrum is discrete.

5.B Eigenvalues of the Hamiltonian

Let us redefine the variables to make them dimensionless:

$$\hat{X} := \sqrt{\frac{m\omega}{\hbar}} X, \quad \hat{P} := \frac{1}{\sqrt{m\hbar\omega}} P,$$

so that

$$[\hat{X}, \hat{P}] = i, \quad H = \hbar\omega \hat{H}, \quad \hat{H} = \frac{1}{2}(\hat{X}^2 + \hat{P}^2).$$

Let us now set

$$a := \frac{1}{\sqrt{2}}(\hat{X} + i\hat{P}), \quad a^\dagger = \frac{1}{\sqrt{2}}(\hat{X} - i\hat{P}),$$

so that

$$\hat{X} = \frac{1}{\sqrt{2}}(a + a^\dagger), \quad \hat{P} = \frac{i}{\sqrt{2}}(a^\dagger - a), \quad [a, a^\dagger] = 1.$$

Then

$$\hat{H} = \frac{1}{2}(\hat{X} - i\hat{P})(\hat{X} + i\hat{P}) + \frac{1}{2} = a^\dagger a + \frac{1}{2}.$$

Defining the **number operator**

$$N := a^\dagger a, \quad \hat{H} = N + \frac{1}{2}.$$

Notice that

$$[N, a] = [a^\dagger a, a] = -a, \quad [N, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger.$$

If $N|\varphi_\nu^i\rangle = \nu|\varphi_\nu^i\rangle$, then

$$H|\varphi_\nu^i\rangle = \hbar\omega \left(\nu + \frac{1}{2} \right) |\varphi_\nu^i\rangle, \quad E = \hbar\omega \left(\nu + \frac{1}{2} \right),$$

where i labels possible degeneracies.

Determination of the spectrum.

- $\nu \geq 0$:

$$0 \leq \|a|\varphi_\nu^i\rangle\|^2 = \langle\varphi_\nu^i|a^\dagger a|\varphi_\nu^i\rangle = \langle\varphi_\nu^i|N|\varphi_\nu^i\rangle = \nu \langle\varphi_\nu^i|\varphi_\nu^i\rangle = \nu.$$

- If $a|\varphi_\nu^i\rangle = 0$, then $\nu = 0$: applying $a^\dagger$, $N|\varphi_\nu^i\rangle = 0 = \nu|\varphi_\nu^i\rangle$. Conversely, for $\nu = 0$ the norm above vanishes, so $a|\varphi_0^i\rangle = 0$.
- a lowers ν by one: from $[N, a] = -a$,

$$N(a|\varphi_\nu^i\rangle) = (aN - a)|\varphi_\nu^i\rangle = (\nu - 1)(a|\varphi_\nu^i\rangle) \implies a|\varphi_\nu^i\rangle \propto |\varphi_{\nu-1}^i\rangle.$$

- $a^\dagger$ raises ν by one: from $[N, a^\dagger] = a^\dagger$,

$$N(a^\dagger|\varphi_\nu^i\rangle) = (a^\dagger N + a^\dagger)|\varphi_\nu^i\rangle = (\nu + 1)(a^\dagger|\varphi_\nu^i\rangle) \implies a^\dagger|\varphi_\nu^i\rangle \propto |\varphi_{\nu+1}^i\rangle.$$

- ν must be a non-negative integer, $\nu \in \mathbb{Z}_0^+$. Otherwise, applying a repeatedly we would reach an eigenvector with $0 < \nu < 1$ whose image under a is non-zero and has negative eigenvalue $\nu - 1 < 0$, contradicting $\nu \geq 0$. For integer ν the chain stops at $a|\varphi_0^i\rangle = 0$.

Therefore

$$N|\varphi_n^i\rangle = n|\varphi_n^i\rangle, \quad n \in \mathbb{Z}_0^+, \quad E_n = \hbar\omega \left(n + \frac{1}{2}\right).$$

Ground state. With the operator a we can obtain the ground state immediately:

$$a|\varphi_0^i\rangle = 0 \implies \frac{1}{\sqrt{2}} \left(\sqrt{\frac{m\omega}{\hbar}} X + \frac{i}{\sqrt{m\hbar\omega}} P \right) |\varphi_0^i\rangle = 0.$$

With $\langle x|\varphi_0^i\rangle = \varphi_0^i(x)$,

$$\left(\frac{m\omega}{\hbar} x + \frac{d}{dx} \right) \varphi_0^i(x) = 0 \implies \varphi_0^i(x) = c e^{-\frac{1}{2} \frac{m\omega}{\hbar} x^2}.$$

The ground state is not degenerate, and neither are the higher energy states (they are obtained from it with $a^\dagger$):

$$\varphi_0(x) = c e^{-\frac{1}{2} \frac{m\omega}{\hbar} x^2}.$$

5.C Eigenstates of the Hamiltonian

Recall that $a^\dagger|\varphi_n\rangle = c|\varphi_{n+1}\rangle$. Then

$$|c|^2 = \|a^\dagger|\varphi_n\rangle\|^2 = \langle\varphi_n|aa^\dagger|\varphi_n\rangle = \langle\varphi_n|a^\dagger a + 1|\varphi_n\rangle = n + 1,$$

so, choosing c real and positive,

$$a^\dagger|\varphi_n\rangle = \sqrt{n+1}|\varphi_{n+1}\rangle \iff a^\dagger|\varphi_{n-1}\rangle = \sqrt{n}|\varphi_n\rangle,$$

and by induction

$$|\varphi_n\rangle = \frac{1}{\sqrt{n!}} (a^\dagger)^n |\varphi_0\rangle.$$

$\{|\varphi_n\rangle\}$ is a basis:

$$\langle\varphi_n|\varphi_m\rangle = \delta_{nm}, \quad \sum_n |\varphi_n\rangle\langle\varphi_n| = \mathbb{1}.$$

Action of the operators.

$$a^\dagger |\varphi_n\rangle = \sqrt{n+1} |\varphi_{n+1}\rangle, \quad a |\varphi_n\rangle = \sqrt{n} |\varphi_{n-1}\rangle,$$

so the matrix elements $\langle \varphi_m | a | \varphi_n \rangle = \sqrt{n} \delta_{m,n-1}$ and $\langle \varphi_m | a^\dagger | \varphi_n \rangle = \sqrt{n+1} \delta_{m,n+1}$ are

$$a = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & \cdots \\ \vdots & & & \ddots & \ddots \end{pmatrix}, \quad a^\dagger = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ \sqrt{1} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & \ddots \\ \vdots & & & \ddots \end{pmatrix}.$$

Wave functions. Since $(a^\dagger)^n |\varphi_0\rangle = \sqrt{n!} |\varphi_n\rangle$,

$$\varphi_n(x) = \frac{1}{\sqrt{n!}} \langle x | (a^\dagger)^n | \varphi_0 \rangle = \frac{1}{\sqrt{n!}} \frac{1}{\sqrt{2^n}} \left[\sqrt{\frac{m\omega}{\hbar}} x - \sqrt{\frac{\hbar}{m\omega}} \frac{d}{dx} \right]^n \varphi_0(x).$$

6 General Properties of Angular Momentum in Quantum Mechanics

6.B Commutation Relations Characteristic of Angular Momentum

The classical orbital angular momentum $L_x = yp_z - zp_y$ is quantised as

$$L_x = YP_z - ZP_y,$$

and similarly for L_y, L_z . Then

$$\begin{aligned} [L_x, L_y] &= [YP_z - ZP_y, ZP_x - XP_z] = [YP_z, ZP_x] + [ZP_y, XP_z] \\ &= Y \underbrace{[P_z, Z]}_{-i\hbar} P_x + X \underbrace{[Z, P_z]}_{i\hbar} P_y = i\hbar (XP_y - YP_x) = i\hbar L_z, \end{aligned}$$

and in general

$$[L_j, L_k] = i\hbar \varepsilon_{jkl} L_l,$$

which has a geometric origin (it reflects the structure of rotations).

Let us define an **angular momentum** $\vec{J}$ as any set of three observables J_x, J_y, J_z that satisfies

$$[J_l, J_m] = i\hbar \varepsilon_{lmn} J_n.$$

Its components are not simultaneously diagonalisable.

Let us introduce the operator

$$\vec{J}^2 := J_x^2 + J_y^2 + J_z^2.$$

Then

$$[\vec{J}^2, \vec{J}] = 0, \quad \text{in particular} \quad [\vec{J}^2, J_z] = 0,$$

so $\vec{J}^2$ and J_z are simultaneously diagonalisable. For example,

$$\begin{aligned} [\vec{J}^2, J_x] &= [J_y^2, J_x] + [J_z^2, J_x] = J_y[J_y, J_x] + [J_y, J_x]J_y + J_z[J_z, J_x] + [J_z, J_x]J_z \\ &= i\hbar[-J_yJ_z - J_zJ_y + J_zJ_y + J_yJ_z] = 0. \end{aligned}$$

6.C General Theory of Angular Momentum

Definitions and notation.

$$J_{\pm} := J_x \pm iJ_y,$$

$$[J_z, J_{\pm}] = \pm \hbar J_{\pm}, \quad [J_+, J_-] = 2\hbar J_z, \quad [\vec{J}^2, J_{\pm}] = [\vec{J}^2, J_z] = 0.$$

Moreover,

$$J_+ J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 + J_y^2 - i[J_x, J_y] = \vec{J}^2 - J_z^2 + \hbar J_z,$$

and similarly $J_- J_+ = \vec{J}^2 - J_z^2 - \hbar J_z$.

Notice that

$$\langle \psi | \vec{J}^2 | \psi \rangle = \sum_i \langle \psi | J_i^2 | \psi \rangle = \sum_i \|J_i | \psi \rangle\|^2 \geq 0,$$

so all eigenvalues of $\vec{J}^2$ are ≥ 0 . We use the notation

Operator	Eigenvalue
$\vec{J}^2$	$j(j+1)\hbar^2, \quad j \geq 0$
J_z	$m\hbar$

so that

$$\vec{J}^2 |k, j, m\rangle = j(j+1)\hbar^2 |k, j, m\rangle, \quad J_z |k, j, m\rangle = m\hbar |k, j, m\rangle,$$

where k is an extra index (the eigenvalues of further observables) such that the labels $\{k, j, m\}$ correspond to a C.S.C.O.

Lemma I. $-j \leq m \leq j$.

Proof.

$$\begin{aligned} \|J_+ |k, j, m\rangle\|^2 &= \langle k, j, m | J_- J_+ |k, j, m\rangle = \langle k, j, m | \vec{J}^2 - J_z^2 - \hbar J_z |k, j, m\rangle \\ &= \hbar^2 [j(j+1) - m^2 - m] \geq 0, \\ \|J_- |k, j, m\rangle\|^2 &= \langle k, j, m | J_+ J_- |k, j, m\rangle = \hbar^2 [j(j+1) - m^2 + m] \geq 0, \end{aligned}$$

that is,

$$\begin{aligned} j(j+1) - m(m+1) &= (j-m)(j+m+1) \geq 0, \\ j(j+1) - m(m-1) &= (j+m)(j-m+1) \geq 0, \end{aligned}$$

which give $-(j+1) \leq m \leq j$ and $-j \leq m \leq j+1$, hence $-j \leq m \leq j$. ■

Lemma II (J_-).

(i) $J_- |k, j, m\rangle = 0 \iff m = -j$. Indeed,

$$J_+ J_- |k, j, m\rangle = \hbar^2 [j(j+1) - m(m-1)] |k, j, m\rangle = \hbar^2 (j+m)(j-m+1) |k, j, m\rangle,$$

which vanishes for $-j \leq m \leq j$ only if $m = -j$. Conversely, for $m = -j$ the norm $\|J_- |k, j, -j\rangle\|^2$ vanishes.

(ii) If $m > -j$, $J_- |k, j, m\rangle \propto |k, j, m-1\rangle$. Since $[\vec{J}^2, J_-] = 0$,

$$\vec{J}^2 (J_- |k, j, m\rangle) = \hbar^2 j(j+1) (J_- |k, j, m\rangle),$$

and since $[J_z, J_-] = -\hbar J_-$,

$$J_z J_- |k, j, m\rangle = J_- J_z |k, j, m\rangle - \hbar J_- |k, j, m\rangle = (m-1)\hbar J_- |k, j, m\rangle.$$

Lemma III (J_+). Analogously,

$$J_+ |k, j, m\rangle = 0 \iff m = j, \\ \vec{J}^2 J_+ |k, j, m\rangle = \hbar^2 j(j+1) J_+ |k, j, m\rangle, \quad J_+ |k, j, m\rangle \propto |k, j, m+1\rangle.$$

Summary.

$$J_\pm |k, j, m\rangle = 0 \iff m = \pm j, \\ \vec{J}^2 J_\pm |k, j, m\rangle = \hbar^2 j(j+1) J_\pm |k, j, m\rangle, \quad J_\pm |k, j, m\rangle \propto |k, j, m \pm 1\rangle.$$

By p (resp. q) successive applications of J_+ (resp. J_-), with p, q integers, we must reach the ends of the ladder:

$$m + p = j, \quad m - q = -j \implies p + q = 2j.$$

Hence j must be an integer or half an integer, and $2j + 1$ values of m are permitted.

$\{|k, j, m\rangle\}$ is a basis:

$$\langle k', j', m' | k, j, m \rangle = \delta_{kk'} \delta_{jj'} \delta_{mm'}, \quad \sum_j \sum_{m=-j}^j \sum_{k=1}^{g(j)} |k, j, m\rangle \langle k, j, m| = \mathbb{1}.$$

Representation.

$$\vec{J}^2 |k, j, m\rangle = \hbar^2 j(j+1) |k, j, m\rangle, \\ J_z |k, j, m\rangle = \hbar m |k, j, m\rangle, \\ J_\pm |k, j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |k, j, m \pm 1\rangle.$$

For $j = 1/2$, $m \in \{-1/2, 1/2\}$, and in the basis $\{|+1/2\rangle, |-1/2\rangle\}$ (e.g. for J_+ acting on $m = -1/2$: $j(j+1) - m(m+1) = \frac{3}{4} + \frac{1}{4} = 1$),

$$J_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} := \frac{\hbar}{2} \sigma_z, \quad J_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \\ J_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} := \frac{\hbar}{2} \sigma_x, \quad J_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} := \frac{\hbar}{2} \sigma_y,$$

where $\{\sigma_x, \sigma_y, \sigma_z\}$ are the **Pauli matrices**.

6.D Application to Orbital Angular Momentum

In the $\{|\vec{r}\rangle\}$ representation, $X_i = x_i$ and $\vec{P} = -i\hbar\nabla$, so

$$L_x = -i\hbar(y\partial_z - z\partial_y), \\ L_y = -i\hbar(z\partial_x - x\partial_z), \\ L_z = -i\hbar(x\partial_y - y\partial_x).$$

In spherical coordinates, $x = r \sin \theta \cos \varphi$, $y = r \sin \theta \sin \varphi$, $z = r \cos \theta$,

$$L_x = i\hbar \left(\sin \varphi \frac{\partial}{\partial \theta} + \frac{\cos \varphi}{\tan \theta} \frac{\partial}{\partial \varphi} \right), \\ L_y = i\hbar \left(-\cos \varphi \frac{\partial}{\partial \theta} + \frac{\sin \varphi}{\tan \theta} \frac{\partial}{\partial \varphi} \right), \\ L_z = -i\hbar \frac{\partial}{\partial \varphi},$$

and therefore

$$\vec{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \frac{1}{\tan \theta} \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right), \quad L_{\pm} = \hbar e^{\pm i \varphi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right). \quad (6.1)$$

The eigenvalue equations become

$$\begin{aligned} -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \frac{1}{\tan \theta} \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \psi(r, \theta, \varphi) &= \hbar^2 l(l+1) \psi(r, \theta, \varphi), \\ -i \frac{\partial}{\partial \varphi} \psi(r, \theta, \varphi) &= m \psi(r, \theta, \varphi), \end{aligned}$$

whose angular solutions are the **spherical harmonics** $Y_l^m(\theta, \varphi)$:

$$\vec{L}^2 Y_l^m(\theta, \varphi) = \hbar^2 l(l+1) Y_l^m(\theta, \varphi), \quad L_z Y_l^m(\theta, \varphi) = \hbar m Y_l^m(\theta, \varphi).$$

Notice that

$$-i \frac{\partial}{\partial \varphi} Y_l^m(\theta, \varphi) = m Y_l^m(\theta, \varphi) \implies Y_l^m(\theta, \varphi) = F_l^m(\theta) e^{im\varphi},$$

and single-valuedness, $Y_l^m(\theta, 0) = Y_l^m(\theta, 2\pi)$, holds if and only if $m \in \mathbb{Z}$, which implies $l \in \mathbb{Z}$. For orbital angular momentum, l and m cannot take half-integer values.

Charged particle in a vector potential. In one dimension, with Lagrangian

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 + q \dot{x} A,$$

the canonical momentum is

$$p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} + q A \implies \dot{x} = \frac{p - q A}{m},$$

and the Hamiltonian

$$\begin{aligned} H = p \dot{x} - \mathcal{L} &= \frac{p(p - q A)}{m} - \frac{(p - q A)^2}{2m} - \frac{q A(p - q A)}{m} \\ &= (p - q A) \left[\frac{p}{m} - \frac{p}{2m} + \frac{q A}{2m} - \frac{q A}{m} \right] = (p - q A) \frac{p - q A}{2m}, \end{aligned}$$

that is,

$$H = \frac{1}{2m} (p - q A)^2.$$

7 Particle in a Central Potential. The Hydrogen Atom

7.A Stationary States of a Particle in a Central Potential

The quantum mechanical Hamiltonian. For a particle of mass μ in a central potential $V(r)$,

$$\left[-\frac{\hbar^2}{2\mu} \Delta + V(r) \right] \psi(\vec{r}) = E \psi(\vec{r}).$$

In spherical coordinates the Laplacian is

$$\Delta = \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{r^2} \left(\frac{\partial^2}{\partial \theta^2} + \frac{1}{\tan \theta} \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) = \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{1}{\hbar^2 r^2} \vec{L}^2,$$

using (6.1). Hence

$$H = -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \underbrace{\frac{1}{2\mu r^2} \vec{L}^2 + V(r)}_{V_{\text{eff}}},$$

and

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{2\mu r^2} \vec{L}^2 + V(r) \right] \psi(r, \theta, \varphi) = E \psi(r, \theta, \varphi).$$

Separation of variables. Notice that $[H, \vec{L}] = 0$: all three components of the angular momentum are conserved quantities, but they do not commute with each other. $\{H, \vec{L}^2, L_z\}$ form a C.S.C.O., so there is a basis of functions $\psi(r, \theta, \varphi)$ such that

$$\begin{aligned} H\psi(r, \theta, \varphi) &= E \psi(r, \theta, \varphi), \\ \vec{L}^2\psi(r, \theta, \varphi) &= \hbar^2 l(l+1) \psi(r, \theta, \varphi), \\ L_z\psi(r, \theta, \varphi) &= \hbar m \psi(r, \theta, \varphi), \end{aligned}$$

of the form $\psi(r, \theta, \varphi) = R(r) Y_l^m(\theta, \varphi)$. Then

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{d^2}{dr^2} r + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right] R(r) = E R(r).$$

Writing

$$R(r) \rightarrow R_{k,l}(r) = \frac{u_{k,l}(r)}{r},$$

with R regular as $r \rightarrow 0$, we get the **radial equation**

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right] u_{k,l}(r) = E_{k,l} u_{k,l}(r), \quad (7.1)$$

where the **centrifugal potential** $l(l+1)\hbar^2/2\mu r^2$ gives a repulsive V_{eff} at short distances.

Behaviour at the origin. Assuming $V(r)$ is less singular than $1/r^2$ at the origin, try $u_{k,l}(r) \propto r^s$ as $r \rightarrow 0$. The dominant terms of (7.1) give

$$-s(s-1) + l(l+1) = 0 \implies s = l+1 \text{ or } s = -l.$$

The solution $u_{k,l} \propto r^{-l}$ is not acceptable at $r = 0$ (for $l = 0$, $R = 1/r$ does not satisfy the Schrödinger equation at the origin, since $\Delta(1/r) = -4\pi\delta(\vec{r})$), whereas $u_{k,l} \propto r^{l+1}$ is. Hence the additional condition to the eigenvalue equation

$$u_{k,l}(0) = 0.$$

7.C The Hydrogen Atom

Quantum mechanical theory of the hydrogen atom. The Schrödinger equation (Gaussian units, $e^2 \equiv q_e^2/4\pi\epsilon_0$) is

$$\left[-\frac{\hbar^2}{2\mu} \Delta - \frac{e^2}{r} \right] \psi(\vec{r}) = E \psi(\vec{r}), \quad \psi_{k,l,m}(\vec{r}) = \frac{1}{r} u_{k,l}(r) Y_l^m(\theta, \varphi),$$

so

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{e^2}{r} \right] u_{k,l}(r) = E_{k,l} u_{k,l}(r), \quad u_{k,l}(0) = 0.$$

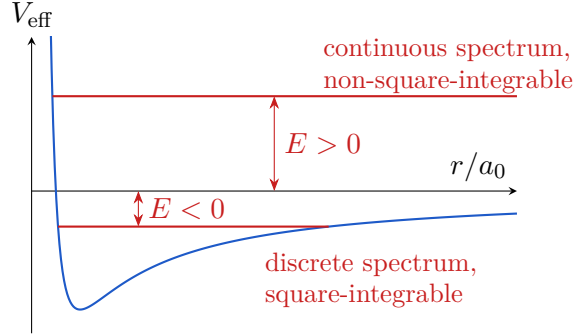


Figure 7.1: Effective potential of the hydrogen atom. Energies $E < 0$ give bound states (discrete spectrum); $E > 0$ gives the continuum.

Define

$$\rho := \frac{r}{a_0}, \quad a_0 := \frac{\hbar^2}{\mu e^2}, \quad \lambda_{k,l} := \sqrt{-\frac{E_{k,l}}{E_I}}, \quad E_I := \frac{\mu e^4}{2\hbar^2}.$$

Then

$$\left[\frac{d^2}{d\rho^2} - \frac{l(l+1)}{\rho^2} + \frac{2}{\rho} - \lambda_{k,l}^2 \right] u_{k,l}(\rho) = 0.$$

Writing

$$u_{k,l}(\rho) = e^{-\rho \lambda_{k,l}} y_{k,l}(\rho), \quad y_{k,l}(0) = 0, \quad y_{k,l}(\rho) = \rho^s \sum_{q=0}^{\infty} c_q \rho^q,$$

the coefficient of ρ^{s-2} (the lowest term) must vanish, which gives $s = l + 1$ or $s = -l$; the acceptable solution is $s = l + 1$.

In general,

$$q(q + 2l + 1) c_q = 2[(q + l)\lambda_{k,l} - 1] c_{q-1}.$$

If the series were infinite, $c_q/c_{q-1} \sim 2\lambda_{k,l}/q$ for large q , so $y_{k,l} \propto e^{2\lambda_{k,l}\rho}$ and $u_{k,l}$ would diverge. We need a finite series, so we cut it off at $q = k$, $k \geq 1$ ($c_k = 0$), where

$$\lambda_{k,l} = \frac{1}{k+l} \implies E_{k,l} = -\frac{E_I}{(k+l)^2}, \quad k \in \mathbb{N}.$$

The coefficients follow from

$$c_q = -\frac{2(k-q)}{q(q+2l+1)(k+l)} c_{q-1} \implies c_q = (-1)^q \left(\frac{2}{k+l} \right)^q \frac{(k-1)!}{(k-q-1)!} \frac{(2l+1)!}{q!(q+2l+1)!} c_0.$$

With the **principal quantum number** $n := k + l$,

$$E_n = -\frac{E_I}{n^2}, \quad n \geq 1.$$

Degeneracy. Fixing n , the allowed pairs are

k	1	2	$\dots$	n
l	$n-1$	$n-2$	$\dots$	0

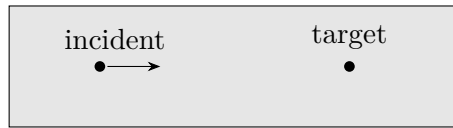
and for each l there are $2l + 1$ possible values of m . Therefore

$$g_n = \sum_{l=0}^{n-1} (2l + 1) = 2 \frac{(n-1)n}{2} + n = n^2.$$

8 An Elementary Approach to the Quantum Theory of Scattering by a Potential

8.A Introduction

In a **scattering** process the final and initial states are composed of the same particles. The scattering is **elastic** when none of the particles' internal states change during the collision.



Simplifying hypotheses:

- incident and target particles have no spin;
- the internal structure of the particles is not taken into account;
- the target is thin enough to enable us to neglect multiple scattering processes;
- we neglect any possibility of coherence between the waves scattered by the different particles that make up the target;
- the interaction between incident and target particles can be described by a potential energy $V(\vec{r}_I - \vec{r}_T)$.

The problem then reduces to that of a single particle of reduced mass μ ,

$$\frac{1}{\mu} = \frac{1}{m_I} + \frac{1}{m_T},$$

in a central potential $V(r)$, where r is the relative distance.

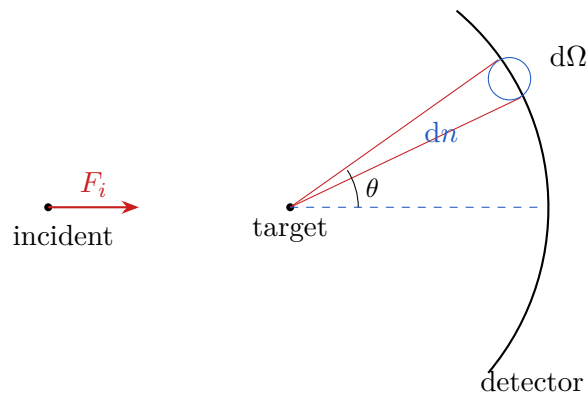


Figure 8.1: Particles scattered into the solid angle $d\Omega(\theta, \varphi)$ are counted by a detector: dn is their number per unit time.

With F_i the flux of incident particles and dn the number of particles scattered into $d\Omega(\theta, \varphi)$ per unit time,

$$dn = F_i \sigma(\theta, \varphi) d\Omega,$$

where $\sigma(\theta, \varphi)$ is the **scattering cross section**, with dimensions of a surface, $[\sigma] = L^2$ (1 barn = 10^{-24} cm^2). The **total scattering cross section** is

$$\sigma := \int d\Omega \sigma(\theta, \varphi).$$

8.B Stationary Scattering States. Calculation of the Cross Section

$$H = H_0 + V(r), \quad H_0 = \frac{\vec{P}^2}{2\mu},$$

$$\psi(\vec{r}, t) = \varphi(\vec{r}) e^{-iEt/\hbar} \implies \left[-\frac{\hbar^2}{2\mu} \Delta + V(r) \right] \varphi(\vec{r}) = E \varphi(\vec{r}),$$

where $V(r)$ is assumed to decrease faster than $1/r$ (which excludes the Coulomb potential).

For a free state,

$$-\frac{\hbar^2}{2\mu} \Delta \varphi_f = E \varphi_f, \quad \varphi_f = A e^{i\vec{k} \cdot \vec{r}} \implies E = \frac{\hbar^2 k^2}{2\mu}.$$

For a scattering state we demand $E = \hbar^2 k^2 / 2\mu$ and write $V(r) = \frac{\hbar^2}{2\mu} U(r)$, so that

$$(\Delta + k^2 - U(r)) \varphi(\vec{r}) = 0.$$

Asymptotic form of stationary scattering states.

$$v_k^{(s)}(\vec{r}) \underset{r \rightarrow \infty}{\sim} e^{ikz} + f_k(\theta, \varphi) \frac{e^{ikr}}{r},$$

the sum of the **transmitted wave** e^{ikz} and a scattered wave with **scattering amplitude** $f_k(\theta, \varphi)$. Note that $(\Delta + k^2) e^{ikr}/r = 0$ for $r > 0$.

Calculation of the cross section using probability currents. A stationary scattering state describes a probability fluid in steady flow, with current

$$\vec{J}(\vec{r}) = \frac{1}{\mu} \text{Re} \left[\varphi^*(\vec{r}) \frac{\hbar}{i} \nabla \varphi(\vec{r}) \right].$$

For the incident plane wave, $|\vec{J}_i| = \hbar k / \mu$. For the scattered wave,

$$(J_d)_r = \frac{\hbar k}{\mu r^2} |f_k(\theta, \varphi)|^2,$$

$$(J_d)_\theta = \frac{\hbar}{\mu r^3} \text{Re} \left[\frac{1}{i} f_k^*(\theta, \varphi) \partial_\theta f_k(\theta, \varphi) \right],$$

$$(J_d)_\varphi = \frac{\hbar}{\mu r^3 \sin \theta} \text{Re} \left[\frac{1}{i} f_k^*(\theta, \varphi) \partial_\varphi f_k(\theta, \varphi) \right].$$

Since $r \rightarrow \infty$, $(J_d)_\theta$ and $(J_d)_\varphi$ (which go as $1/r^3$) are negligible compared to $(J_d)_r$ (which goes as $1/r^2$). Then, with C a constant,

$$F_i = C |\vec{J}_i| = C \frac{\hbar k}{\mu}, \quad dn = C \vec{J}_d \cdot d\vec{S} = C (J_d)_r r^2 d\Omega = C \frac{\hbar k}{\mu} |f_k(\theta, \varphi)|^2 d\Omega,$$

and therefore

$$\sigma(\theta, \varphi) = |f_k(\theta, \varphi)|^2.$$

Integral scattering equation. Write the equation as $(\Delta + k^2)\varphi(\vec{r}) = U(\vec{r})\varphi(\vec{r})$ and let G be a Green's function, $(\Delta + k^2)G(\vec{r}) = \delta(\vec{r})$. Then

$$\varphi(\vec{r}) = \varphi_0(\vec{r}) + \int d^3r' G(\vec{r} - \vec{r}') U(\vec{r}') \varphi(\vec{r}'),$$

with φ_0 a solution of the free equation, and

$$G_{\pm}(\vec{r}) = -\frac{1}{4\pi} \frac{e^{\pm ikr}}{r}.$$

The outgoing condition selects G_+ :

$$v_k^{(s)}(\vec{r}) = e^{ikz} + \int d^3r' G_+(\vec{r} - \vec{r}') U(\vec{r}') v_k^{(s)}(\vec{r}'). \quad (8.1)$$

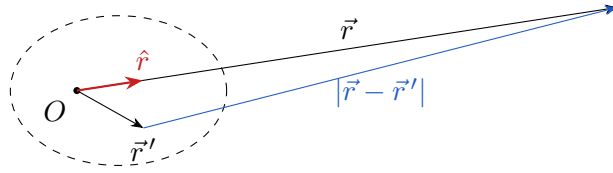


Figure 8.2: Far from the scattering region, $|\vec{r} - \vec{r}'| \simeq r - \hat{r} \cdot \vec{r}'$.

For $r \rightarrow \infty$, $|\vec{r} - \vec{r}'| \simeq r - \hat{r} \cdot \vec{r}'$, so

$$G_+(\vec{r} - \vec{r}') = -\frac{1}{4\pi} \frac{e^{ik|\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} \underset{r \rightarrow \infty}{\simeq} -\frac{1}{4\pi} \frac{e^{ikr}}{r} e^{-ik\hat{r} \cdot \vec{r}'},$$

$$v_k^{(s)}(\vec{r}) \underset{r \rightarrow \infty}{\simeq} e^{ikz} - \frac{1}{4\pi} \frac{e^{ikr}}{r} \int d^3r' e^{-ik\hat{r} \cdot \vec{r}'} U(\vec{r}') v_k^{(s)}(\vec{r}'),$$

and hence

$$f_k(\theta, \varphi) = -\frac{1}{4\pi} \int d^3r' e^{-ik\hat{r} \cdot \vec{r}'} U(\vec{r}') v_k^{(s)}(\vec{r}').$$

In general we write the incident wave as $e^{ikz} \rightarrow e^{i\vec{k}_i \cdot \vec{r}}$ and define the scattered wave vector $\vec{k}_s := k\hat{r}$.

The Born approximation. Replacing $v_k^{(s)}$ inside the integral of (8.1) by the whole right-hand side,

$$\begin{aligned} v_k^{(s)}(\vec{r}) &= e^{i\vec{k}_i \cdot \vec{r}} + \underbrace{\int d^3r' G_+(\vec{r} - \vec{r}') U(\vec{r}') e^{i\vec{k}_i \cdot \vec{r}'} }_{\text{first order}} \\ &\quad + \int d^3r' \int d^3r'' G_+(\vec{r} - \vec{r}') U(\vec{r}') G_+(\vec{r}' - \vec{r}'') U(\vec{r}'') v_k^{(s)}(\vec{r}''). \end{aligned}$$

The process can be repeated indefinitely. At first order, $v_k^{(s)}(\vec{r}') \simeq e^{i\vec{k}_i \cdot \vec{r}'}$, and

$$f_k(\theta, \varphi) = -\frac{1}{4\pi} \int d^3r' \underbrace{e^{-ik\hat{r} \cdot \vec{r}'}}_{e^{-i\vec{k}_s \cdot \vec{r}'}} U(\vec{r}') e^{i\vec{k}_i \cdot \vec{r}'} = -\frac{1}{4\pi} \int d^3r' e^{-i(\vec{k}_s - \vec{k}_i) \cdot \vec{r}'} U(\vec{r}').$$

With the **scattering wave vector** $\vec{K} := \vec{k}_s - \vec{k}_i$, the **Born approximation** is

$$f_k^{(B)}(\theta, \varphi) = -\frac{1}{4\pi} \int d^3r' e^{-i\vec{K} \cdot \vec{r}'} U(\vec{r}').$$

The scattering amplitude is proportional to the Fourier transform of the potential energy:

$$\sigma_k^{(B)}(\theta, \varphi) = \frac{\mu^2}{4\pi^2 \hbar^4} \left| \int d^3r e^{-i\vec{K} \cdot \vec{r}} V(\vec{r}) \right|^2.$$

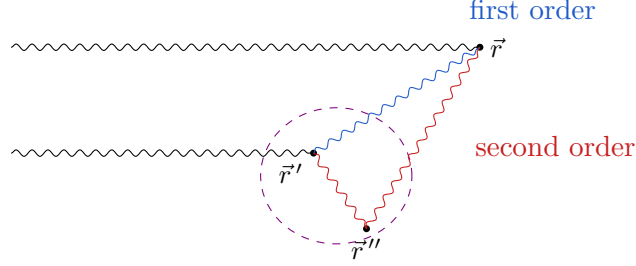


Figure 8.3: First- and second-order terms of the Born series: the incident wave is scattered once, or twice, inside the region where the potential acts.

9 Electron Spin

9.C Non-Relativistic Description of a Spin-1/2 Particle

Different sets of C.S.C.O. are, for example,

$$\{X, Y, Z, \vec{S}^2, S_z\}, \quad \{P_x, P_y, P_z, \vec{S}^2, S_z\}, \quad \{H, \vec{L}^2, L_z, \vec{S}^2, S_z\}.$$

The state space is

$$\mathcal{E} = \mathcal{E}_{\vec{r}} \otimes \mathcal{E}_s.$$

In the $\{|\vec{r}\rangle \otimes |\varepsilon\rangle\}$ basis, with $\varepsilon = \pm$,

$$\langle \vec{r}', \varepsilon' | \vec{r}, \varepsilon \rangle = \delta_{\varepsilon' \varepsilon} \delta(\vec{r}' - \vec{r}),$$

$$\sum_{\varepsilon} \int d^3r |\vec{r}, \varepsilon\rangle \langle \vec{r}, \varepsilon| = \int d^3r (|\vec{r}, +\rangle \langle \vec{r}, +| + |\vec{r}, -\rangle \langle \vec{r}, -|) = \mathbb{1},$$

so that

$$|\psi\rangle = \sum_{\varepsilon} \int d^3r |\vec{r}, \varepsilon\rangle \langle \vec{r}, \varepsilon | \psi \rangle, \quad \psi_+(\vec{r}) = \langle \vec{r}, + | \psi \rangle, \quad \psi_-(\vec{r}) = \langle \vec{r}, - | \psi \rangle.$$

We need two functions to characterise completely the state of a spin-1/2 particle. They are gathered in the two-component **spinor**

$$[\psi](\vec{r}) := \begin{pmatrix} \psi_+(\vec{r}) \\ \psi_-(\vec{r}) \end{pmatrix}, \quad [\psi]^\dagger(\vec{r}) = \begin{pmatrix} \psi_+^*(\vec{r}) & \psi_-^*(\vec{r}) \end{pmatrix}.$$

Notice that if $|\psi\rangle = |\varphi\rangle \otimes |\chi\rangle$, with $|\varphi\rangle = \int d^3r \varphi(\vec{r}) |\vec{r}\rangle$ and $|\chi\rangle = c_+ |+\rangle + c_- |-\rangle$, then

$$[\psi](\vec{r}) = \begin{pmatrix} \varphi(\vec{r}) c_+ \\ \varphi(\vec{r}) c_- \end{pmatrix} = \varphi(\vec{r}) \begin{pmatrix} c_+ \\ c_- \end{pmatrix}.$$

10 Addition of Angular Momenta

10.C Statement of the Problem

Consider a physical system formed by the union of systems 1 and 2, so that $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$, with ($i = 1, 2$)

$$\begin{aligned}\vec{J}_i^2 |k_i, j_i, m_i\rangle &= \hbar^2 j_i(j_i + 1) |k_i, j_i, m_i\rangle, \\ J_{iz} |k_i, j_i, m_i\rangle &= \hbar m_i |k_i, j_i, m_i\rangle, \\ J_{i\pm} |k_i, j_i, m_i\rangle &= \hbar \sqrt{j_i(j_i + 1) - m_i(m_i \pm 1)} |k_i, j_i, m_i \pm 1\rangle.\end{aligned}$$

The **total angular momentum** is

$$\vec{J} = \vec{J}_1 + \vec{J}_2,$$

where $\vec{J}_1$ and $\vec{J}_2$ act on different subspaces. Then

$$[\vec{J}^2, \vec{J}_i^2] = 0, \quad [J_z, \vec{J}_i^2] = 0.$$

Notice that

$$\vec{J}^2 = \vec{J}_1^2 + \vec{J}_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+},$$

so $\vec{J}^2$ does not commute with J_{1z} , J_{2z} . There are two natural C.S.C.O.:

- $\{\vec{J}_1^2, \vec{J}_2^2, J_{1z}, J_{2z}\}$: study of the individual angular momenta;
- $\{\vec{J}_1^2, \vec{J}_2^2, \vec{J}^2, J_z\}$: study of the total angular momentum.

In both, the eigenvalues of $\vec{J}_1^2$ and $\vec{J}_2^2$ do not change.

Eigenvalues of J_z and their degeneracy. Within $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$ we consider the subspace $\mathcal{E}(j_1, j_2)$ of fixed $j_1 \geq j_2$, of dimension $(2j_1 + 1)(2j_2 + 1)$. The vectors $|j_1, j_2; m_1, m_2\rangle$ are already eigenstates of J_z :

$$J_z |j_1, j_2; m_1, m_2\rangle = (J_{1z} + J_{2z}) |j_1, j_2; m_1, m_2\rangle = \hbar(m_1 + m_2) |j_1, j_2; m_1, m_2\rangle.$$

With $M := m_1 + m_2$,

$$M \in \{j_1 + j_2, j_1 + j_2 - 1, \dots, -(j_1 + j_2)\}.$$

On the other hand,

$$J \in \{j_1 + j_2, j_1 + j_2 - 1, \dots, |j_1 - j_2|\},$$

and the common eigenvectors of $\vec{J}^2$ and J_z satisfy

$$\vec{J}^2 |J, M\rangle = \hbar^2 J(J + 1) |J, M\rangle, \quad J_z |J, M\rangle = \hbar M |J, M\rangle.$$

Special case of two spins 1/2. Here $\mathcal{E} = \mathcal{E}(1/2, 1/2)$ and $S \in \{1, 0\}$, with $M \in \{1, 0, -1\}$ for $S = 1$ and $M = 0$ for $S = 0$. In the notation $|S, M\rangle$ and $|m_1 m_2\rangle$:

- $|1, 1\rangle = |++\rangle$.
- Applying $S_- = S_{1-} + S_{2-}$, with $S_- |1, 1\rangle = \hbar\sqrt{2} |1, 0\rangle$:

$$|1, 0\rangle = \frac{1}{\hbar\sqrt{2}} S_- |++\rangle = \frac{1}{\hbar\sqrt{2}} (S_{1-} + S_{2-}) |++\rangle = \frac{1}{\sqrt{2}} [|+-\rangle + |-+\rangle].$$

- $|1, -1\rangle = \frac{1}{\hbar\sqrt{2}} S_- |1, 0\rangle = |--\rangle$.

- $|0, 0\rangle = \alpha |+-\rangle + \beta |-+\rangle$, with

$$\langle 0, 0|0, 0\rangle = 1 \Rightarrow |\alpha|^2 + |\beta|^2 = 1, \quad \langle 1, 0|0, 0\rangle = 0 \Rightarrow \frac{1}{\sqrt{2}}(\alpha + \beta) = 0 \Rightarrow \alpha = -\beta,$$

so that

$$|0, 0\rangle = \frac{1}{\sqrt{2}}[|+-\rangle - |-+\rangle].$$

Remark. The handwritten notes include here a screenshot of the numerical tables of Clebsch–Gordan coefficients $\langle JM(jj')|mm'\rangle$ from W. K. Tung, *Group Theory in Physics* (World Scientific, 1985), for $1/2 \times 1/2$, $1 \times 1/2$, 1×1 , $3/2 \times 1/2$, $2 \times 1/2$, $3/2 \times 1$ and 2×1 . It is not reproduced here; see the reference (or any standard table, e.g. the Particle Data Group review).

11 Stationary Perturbation Theory

11.B Perturbation of a Non-Degenerate Level

Let

$$H(\lambda) = H_0 + \lambda \hat{W}, \quad \lambda \in \mathbb{R}, \lambda \ll 1,$$

with

$$H_0 |\varphi_n^i\rangle = E_n^0 |\varphi_n^i\rangle, \quad \langle \varphi_n^i | \varphi_m^{i'} \rangle = \delta_{nm} \delta_{ii'}, \quad \sum_p \sum_i |\varphi_p^i\rangle \langle \varphi_p^i| = \mathbb{1},$$

and we look for

$$H(\lambda) |\psi(\lambda)\rangle = E(\lambda) |\psi(\lambda)\rangle$$

as power series

$$E(\lambda) = \varepsilon_0 + \lambda \varepsilon_1 + \cdots + \lambda^q \varepsilon_q + \cdots, \\ |\psi(\lambda)\rangle = |0\rangle + \lambda |1\rangle + \cdots + \lambda^q |q\rangle + \cdots,$$

so that

$$(H_0 + \lambda \hat{W}) \left(\sum_{q=0}^{\infty} \lambda^q |q\rangle \right) = \left(\sum_{q'=0}^{\infty} \lambda^{q'} \varepsilon_{q'} \right) \left(\sum_{q=0}^{\infty} \lambda^q |q\rangle \right).$$

Order by order in λ :

$$\begin{aligned} \lambda^0: & H_0 |0\rangle = \varepsilon_0 |0\rangle, \\ \lambda^1: & (H_0 - \varepsilon_0) |1\rangle + (\hat{W} - \varepsilon_1) |0\rangle = 0, \\ \lambda^2: & (H_0 - \varepsilon_0) |2\rangle + (\hat{W} - \varepsilon_1) |1\rangle - \varepsilon_2 |0\rangle = 0, \\ \lambda^q: & (H_0 - \varepsilon_0) |q\rangle + (\hat{W} - \varepsilon_1) |q-1\rangle - \varepsilon_2 |q-2\rangle - \cdots - \varepsilon_q |0\rangle = 0. \end{aligned}$$

We choose $|\psi(\lambda)\rangle$ such that

$$\langle \psi(\lambda) | \psi(\lambda) \rangle = 1, \quad \langle 0 | \psi(\lambda) \rangle \in \mathbb{R}.$$

At zeroth order $\langle 0|0\rangle = 1$, and

$$\langle \psi(\lambda) | \psi(\lambda) \rangle = \langle 0|0\rangle + \lambda [\langle 1|0\rangle + \langle 0|1\rangle] + \mathcal{O}(\lambda^2),$$

which gives

$$\begin{aligned}\langle 0|1\rangle &= \langle 1|0\rangle = 0, \\ \langle 0|2\rangle &= \langle 2|0\rangle = -\frac{1}{2}\langle 1|1\rangle, \\ \langle 0|q\rangle &= \langle q|0\rangle = -\frac{1}{2}[\langle q-1|1\rangle + \langle q-2|2\rangle + \cdots + \langle 1|q-1\rangle].\end{aligned}$$

For the non-degenerate level E_n^0 we take $|0\rangle = |\varphi_n\rangle$ and $\varepsilon_0 = E_n^0$. Projecting the first-order equation onto $\langle \varphi_n|$,

$$\underbrace{(\varepsilon_0 - \varepsilon_0)}_0 \langle \varphi_n|1\rangle + \langle \varphi_n|\hat{W} - \varepsilon_1|\varphi_n\rangle = 0 \quad \implies \quad \varepsilon_1 = \langle \varphi_n|\hat{W}|\varphi_n\rangle.$$

With $W = \lambda\hat{W}$, the **energy correction** is

$$E_n(\lambda) = E_n^0 + \langle \varphi_n|W|\varphi_n\rangle + \mathcal{O}(\lambda^2).$$

Projecting the first-order equation onto $\sum_{p \neq n} \sum_i |\varphi_p^i\rangle \langle \varphi_p^i|$,

$$\begin{aligned}\sum_{p \neq n} \sum_i |\varphi_p^i\rangle \langle \varphi_p^i|1\rangle (E_p^0 - E_n^0) + \sum_{p \neq n} \sum_i |\varphi_p^i\rangle \langle \varphi_p^i|\hat{W}|\varphi_n\rangle &= 0 \\ \implies |1\rangle &= \sum_{p \neq n} \sum_i \frac{\langle \varphi_p^i|\hat{W}|\varphi_n\rangle}{E_n^0 - E_p^0} |\varphi_p^i\rangle\end{aligned}$$

(the component along $|\varphi_n\rangle$ vanishes, $\langle 0|1\rangle = 0$). Hence

$$|\psi_n(\lambda)\rangle = |\varphi_n\rangle + \sum_{p \neq n} \sum_i \frac{\langle \varphi_p^i|W|\varphi_n\rangle}{E_n^0 - E_p^0} |\varphi_p^i\rangle + \mathcal{O}(\lambda^2).$$

13 Approximation Methods for Time-Dependent Problems

13.A Statement of the Problem

At $t = 0$ a perturbation is applied:

$$H(t) = H_0 + W(t), \quad W(t) = \lambda\hat{W}(t), \quad \lambda \ll 1,$$

where W is typically non-diagonal in the eigenbasis of H_0 , $H_0|\varphi_n\rangle = E_n|\varphi_n\rangle$. How does the system evolve?

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = (H_0 + \lambda\hat{W}(t)) |\psi(t)\rangle, \quad |\psi(t=0)\rangle = |\varphi_i\rangle.$$

The quantity of interest is the transition probability

$$\mathcal{P}_{if}(t) = |\langle \varphi_f|\psi(t)\rangle|^2.$$

13.B Approximate Solution of the Schrödinger Equation

Expanding in the eigenbasis of H_0 ,

$$|\psi(t)\rangle = \sum_n c_n(t) |\varphi_n\rangle, \quad c_n(t) = \langle \varphi_n | \psi(t) \rangle,$$

with

$$\hat{W}_{nm} := \langle \varphi_n | \hat{W} | \varphi_m \rangle, \quad \langle \varphi_n | H_0 | \varphi_m \rangle = E_n \delta_{nm}, \quad \sum_n |\varphi_n\rangle\langle \varphi_n| = \mathbb{1},$$

the Schrödinger equation becomes

$$i\hbar \frac{d}{dt} c_n(t) = E_n c_n(t) + \lambda \sum_k \hat{W}_{nk}(t) c_k(t).$$

Defining $b_n(t) = c_n(t) e^{iE_n t/\hbar}$,

$$i\hbar e^{-iE_n t/\hbar} \frac{d}{dt} b_n(t) = \lambda \sum_k \hat{W}_{nk} b_k(t) e^{-iE_k t/\hbar},$$

that is,

$$i\hbar \frac{d}{dt} b_n(t) = \lambda \sum_k \hat{W}_{nk}(t) b_k(t) e^{i\omega_{nk} t}, \quad \omega_{nk} := \frac{E_n - E_k}{\hbar}.$$

Using $b_n(t) = b_n^{(0)}(t) + \lambda b_n^{(1)}(t) + \mathcal{O}(\lambda^2)$:

$$\text{order } 0 : \quad i\hbar \frac{d}{dt} b_n^{(0)}(t) = 0,$$

$$\text{order } r \neq 0 : \quad i\hbar \frac{d}{dt} b_n^{(r)}(t) = \sum_k e^{i\omega_{nk} t} \hat{W}_{nk}(t) b_k^{(r-1)}(t).$$

The initial condition $b_n(t=0) = \delta_{ni}$ gives $b_n^{(0)}(0) = \delta_{ni}$ (so $b_n^{(0)}(t) = \delta_{ni}$) and $b_n^{(r)}(0) = 0$ for $r \geq 1$. For $r = 1$,

$$i\hbar \frac{d}{dt} b_n^{(1)}(t) = \sum_k e^{i\omega_{nk} t} \hat{W}_{nk}(t) \delta_{ki} = e^{i\omega_{ni} t} \hat{W}_{ni}(t) \implies b_n^{(1)}(t) = -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_{ni} t'} \hat{W}_{ni}(t').$$

Then

$$\mathcal{P}_{if}(t) = |\langle \varphi_f | \psi(t) \rangle|^2 = |c_f(t)|^2 = |b_f(t)|^2 = \left| b_f^{(0)}(t) + \lambda b_f^{(1)}(t) + \dots \right|^2.$$

Assuming $|\varphi_i\rangle \neq |\varphi_f\rangle$ and keeping only the first order in λ , $\mathcal{P}_{if} = \lambda^2 \left| b_f^{(1)}(t) \right|^2$, i.e.

$$\mathcal{P}_{if}(t) = \frac{1}{\hbar^2} \left| \int_0^t dt' e^{i\omega_{fi} t'} W_{fi}(t') \right|^2.$$

13.C Coupling with the States of the Continuous Spectrum

If E_f belongs to a continuous part of the spectrum of H_0 , we cannot measure the probability of finding the system in a well-defined state $|\varphi_f\rangle$ at time t : we need to integrate over a group of states, using the density of states.

Simple case. The probability of finding a particle with momentum around $\vec{p}_f$, i.e. in the solid angle $\delta\Omega_f$ and with energy around E_f , is

$$\delta\mathcal{P}(\vec{p}_f, t) = \int_{\vec{p} \in D_f} d^3p |\langle \vec{p} | \psi(t) \rangle|^2.$$

With $d^3p = p^2 dp d\Omega = \rho(E) dE d\Omega$,

$$\delta\mathcal{P}(\vec{p}_f, t) = \int_{\Omega \in \delta\Omega_f, E \in \delta E_f} d\Omega dE \rho(E) |\langle \vec{p} | \psi(t) \rangle|^2.$$

In general. With $\{|\alpha\rangle\}$ a basis of eigenstates of H_0 in the continuum,

$$\delta\mathcal{P}(\alpha_f, t) = \int_{\alpha \in D_f} d\alpha |\langle \alpha | \psi(t) \rangle|^2, \quad d\alpha = \rho(\beta, E) d\beta dE,$$

where β stands for the other quantum numbers which, together with E , form a C.S.C.O. Then

$$\delta\mathcal{P}(\alpha_f, t) = \int_{\beta \in \delta\beta_f, E \in \delta E_f} d\beta dE \rho(\beta, E) |\langle \beta, E | \psi(t) \rangle|^2.$$

For a constant perturbation W switched on at $t = 0$, the first-order result gives

$$|\langle \beta, E | \psi(t) \rangle|^2 = \frac{1}{\hbar^2} |\langle \beta, E | W | \varphi_i \rangle|^2 F\left(t, \frac{E - E_i}{\hbar}\right), \quad F(t, \omega) = \left[\frac{\sin(\omega t/2)}{\omega/2} \right]^2,$$

where it is assumed that the initial state belongs to the discrete part of the spectrum of H_0 . Therefore

$$\delta\mathcal{P}(\varphi_i, \alpha_f, t) = \frac{1}{\hbar^2} \int_{\beta \in \delta\beta_f, E \in \delta E_f} d\beta dE \rho(\beta, E) |\langle \beta, E | W | \varphi_i \rangle|^2 F\left(t, \frac{E - E_i}{\hbar}\right).$$

For $t \rightarrow \infty$,

$$\lim_{t \rightarrow \infty} F\left(t, \frac{E - E_i}{\hbar}\right) = \pi t \delta\left(\frac{E - E_i}{2\hbar}\right) = 2\pi\hbar t \delta(E - E_i).$$

If $\delta\beta_f$ is small and $t \rightarrow \infty$,

$$\delta\mathcal{P}(\varphi_i, \alpha_f, t) = \frac{1}{\hbar^2} \delta\beta_f 2\pi\hbar t \rho(\beta_f, E_f = E_i) |\langle \beta_f, E_f = E_i | W | \varphi_i \rangle|^2.$$

The **transition probability per unit time** is

$$\delta\mathcal{W}(\varphi_i, \alpha_f) := \frac{d}{dt} \delta\mathcal{P}(\varphi_i, \alpha_f, t),$$

and its density per unit β_f , $w(\varphi_i, \alpha_f) := \delta\mathcal{W}(\varphi_i, \alpha_f) / \delta\beta_f$, is **Fermi's golden rule**:

$$w(\varphi_i, \alpha_f) = \frac{2\pi}{\hbar} |\langle \beta_f, E_f = E_i | W | \varphi_i \rangle|^2 \rho(\beta_f, E_f = E_i),$$

when the energy E_i belongs to δE_f , and 0 otherwise.

14 Systems of Identical Particles

Two particles are said to be **identical** if all their intrinsic properties (mass, spin, charge, etc.) are exactly the same: no experiment can distinguish one from the other. For example, all electrons in the universe are identical.

Permutation operators. For two particles,

$$P_{21} |1:u_i; 2:u_j\rangle := |2:u_i; 1:u_j\rangle = |1:u_j; 2:u_i\rangle,$$

with

$$P_{21}^2 = \mathbb{1}, \quad P_{21}^\dagger = P_{21}.$$

Its eigenstates are

$$\begin{aligned} P_{21} |\psi_S\rangle &= + |\psi_S\rangle & |\psi_S\rangle &\text{ symmetric,} \\ P_{21} |\psi_A\rangle &= - |\psi_A\rangle & |\psi_A\rangle &\text{ antisymmetric.} \end{aligned}$$

The **symmetriser** and **antisymmetriser**

$$S := \frac{1}{2}(\mathbb{1} + P_{21}), \quad A := \frac{1}{2}(\mathbb{1} - P_{21})$$

are projectors:

$$S^2 = S, \quad A^2 = A, \quad S^\dagger = S, \quad A^\dagger = A, \quad [S, A] = 0, \quad S + A = \mathbb{1}.$$

Let P_α be an arbitrary permutation operator associated with a system of N particles with the same spin, where α represents an arbitrary permutation of the first N integers. Then

$$\begin{aligned} P_\alpha |\psi_S\rangle &= |\psi_S\rangle & |\psi_S\rangle &\text{ completely symmetric,} \\ P_\alpha |\psi_A\rangle &= \varepsilon_\alpha |\psi_A\rangle & |\psi_A\rangle &\text{ completely antisymmetric,} \end{aligned}$$

with $\varepsilon_\alpha = +1$ if P_α is an even permutation and $\varepsilon_\alpha = -1$ if it is odd, and

$$S := \frac{1}{N!} \sum_{\alpha} P_\alpha, \quad A := \frac{1}{N!} \sum_{\alpha} \varepsilon_\alpha P_\alpha.$$

14.C The Symmetrisation Postulate

When a system includes several identical particles, only certain kets of its state space can describe its physical states. **Physical kets are, depending on the nature of the identical particles, either completely symmetric or completely antisymmetric** with respect to permutation of these particles:

- symmetric: **bosons**;
- antisymmetric: **fermions**.

Pauli's exclusion principle: two identical fermions cannot occupy the same individual quantum state.

Ground state of N identical particles. Bosons all occupy the lowest individual state; fermions fill the individual levels one by one up to the **Fermi energy** of the system (at $T = 0$).

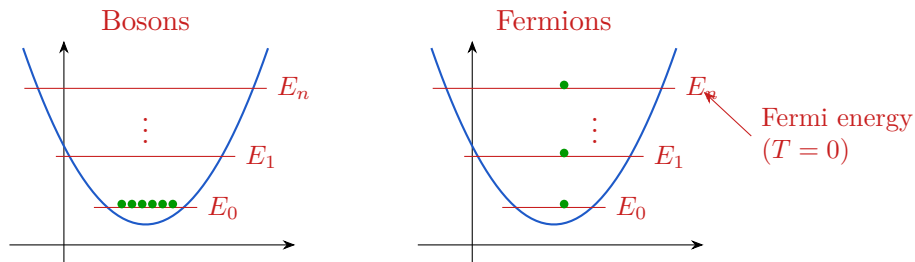


Figure 14.1: Ground state of N identical (spinless) particles in a harmonic well.